Sales & Profit Analysis

End-to-End Data Analysis Capstone Project

Data Analyst Course — Week 4: Data Visualization & Capstone Project

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Certificate / Declaration

This is to certify that the project titled “Sales & Profit Analysis — End-to-End Data Analysis Capstone Project” has been completed by **Parth Garg** as part of the Data Analyst Course, Week 4: Data Visualization & Capstone Project, offered by Axcentra.

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5. Project Overview

This capstone project was completed as the final requirement of Week 4 of the Data Analyst Course, which focused on Data Visualization and end-to-end data analysis. The project follows a complete analytics workflow: starting from a raw business sales dataset, cleaning and preparing the data in Python, analyzing it using Pandas, cross-checking the results in Microsoft Excel, and finally building an interactive dashboard in Tableau to visually communicate business performance.

The dataset used represents a company's sales and profit records across multiple countries, customer segments, and products, spanning September 2013 to December 2014. The analysis focuses on identifying which parts of the business generate the most revenue and profit, and where performance is weaker.

6. Project Objective

The objective of this project is to demonstrate a complete, practical data analysis workflow using the tools introduced in the course. Specifically, the project aims to:

- sample business sales dataset for analysis.
- Analyze sales, profit, and units sold across country, segment, and product dimensions using Python and Pandas.
- Validate and present analysis results in Microsoft Excel.
- Design an interactive dashboard in Tableau to visualize key business metrics.
- Summarize findings into clear, actionable business insights and recommendations.

7. Dataset Description

The dataset (Sample data.xlsx / Sheet1) contains 700 sales transaction records with 16 columns, covering the period from September 2013 to December 2014. The table below describes each field in the dataset.

Column	Description
Segment	Customer segment (e.g., Government, Enterprise, Small Business)
Country	Country where the sale occurred
Product	Product name sold
Discount Band	Discount tier applied (Low / Medium / High / Unknown)
Units Sold	Number of units sold in the transaction
Manufacturing Price	Cost to manufacture one unit
Sale Price	Price per unit sold to the customer
Gross Sales	Revenue before discount
Discounts	Discount amount applied
Sales	Net sales revenue after discount
COGS	Cost of Goods Sold
Profit	Sales minus COGS
Date / Month Number / Month Name / Year	Transaction date fields

Segment	Country	Product	Discount Band	Units Sold	Sale Price	Sales	Profit	Date
Government	Germany	Carretera	Unknown	1513.0	350	529,550	136,170	2014-12-01
Government	Germany	Paseo	Unknown	1006.0	350	352,100	90,540	2014-06-01
Government	Canada	Paseo	Unknown	1725.0	350	603,750	155,250	2013-11-01
Government	Germany	Paseo	Unknown	1513.0	350	529,550	136,170	2014-12-01
Government	Germany	Velo	Unknown	1006.0	350	352,100	90,540	2014-06-01
Government	France	VTT	Unknown	1527.0	350	534,450	137,430	2013-09-01
Government	France	Amarilla	Unknown	2750.0	350	962,500	247,500	2014-02-01
Government	Mexico	Carretera	Low	1210.0	350	419,265	104,665	2014-03-01
Government	Mexico	Carretera	Low	1397.0	350	484,060	120,840	2014-10-01
Government	France	Carretera	Low	2155.0	350	746,708	186,408	2014-12-01

Figure 7.1: Snapshot of the cleaned dataset (first 10 records)

8. Tools and Technologies Used

- Python (Pandas) — used in Jupyter Notebook for data loading, cleaning, and analysis.
- Microsoft Excel — used to review, verify, and present the analysis results generated by Python.
- Tableau (Tableau Public / Desktop) — used to build the interactive “Sales & Profit Dashboard.”

Note: This project used Tableau for visualization, not Power BI. The analysis was performed using Python/Pandas and Excel; SQL and machine learning / forecasting techniques were not part of this project's scope.

9. Data Cleaning and Preparation

Before analysis, the raw dataset was reviewed for missing and inconsistent values using Python/Pandas. The following cleaning steps were carried out:

- The dataset was loaded and checked column by column for missing values.
- The 'Discount Band' column had 53 missing entries out of 700 records. These were filled with the label "Unknown" rather than being dropped, so that no transaction records were lost.
- After cleaning, the dataset (Cleaned_Sample_Data.xlsx) contains 700 complete records across 16 columns with zero missing values.

The table below shows the final distribution of the Discount Band column after cleaning:

Discount Band	Number of Records
High	245
Medium	242
Low	160
Unknown	53

10. Exploratory Data Analysis

An initial exploration of the cleaned dataset showed the following structure:

- 5 customer segments: Government, Small Business, Enterprise, Midmarket, Channel Partners.
- 5 countries: United States of America, Canada, France, Germany, Mexico.
- 6 products: Paseo, VTT, Amarilla, Velo, Montana, Carretera.
- The dataset spans September 2013 to December 2014 — 175 records fall in 2013 (September–December only) and 525 records fall in 2014 (full year). This means 2013 and 2014 are not directly comparable on a like-for-like annual basis.

11. Data Analysis Using Python / Pandas

Using Jupyter Notebook, the cleaned dataset was loaded into a Pandas DataFrame and grouped/aggregated along several dimensions to answer key business questions:

- Total Sales and Profit grouped by Country.
- Total Profit grouped by Product.
- Total Sales grouped by Segment.
- Total Sales grouped by Product.
- Total Sales grouped by Month Number (Monthly Sales).
- Total Sales grouped by Year (Yearly Sales).
- Combined Sales, Profit, Units Sold, and Profit Margin (%) summarized by Product, Country, and Segment.

The results of these Pandas aggregations were exported into a single Excel workbook, Capstone_Analysis_Results.xlsx, for further review and use in Tableau.

12. Excel Analysis

The Excel workbook Capstone_Analysis_Results.xlsx contains 9 sheets, each holding a specific analysis result generated from Python: Sales by Country, Profit by Product, Sales by Segment, Sales by Product, Monthly Sales, Yearly Sales, Product Analysis, Country Analysis, and Segment Analysis. The combined Analysis sheets (Product/Country/Segment Analysis) bring together Sales, Profit, Units Sold, and Profit Margin (%) in one view for each dimension.

Segment	Sales	Profit	Units_Sold	it Margin (%)
overnmer	52504261	11388173	470673.5	21.69
all Busin	42427919	4143169	153139	9.765194
Enterprise	19611694	-614546	168552	-3.133567
Midmarke	2381883	660103.1	172178	27.7135
nnel Parti	1800594	1316803	161263.5	73.13161

Figure 12.1: Segment Analysis sheet from Capstone_Analysis_Results.xlsx (actual output)

12.1 Segment Analysis (from Excel)

Segment	Sales	Profit	Units Sold	Margin %
Government	52,504,261	11,388,173	470,673.5	21.69%
Small Business	42,427,919	4,143,169	153,139.0	9.77%
Enterprise	19,611,694	-614,546	168,552.0	-3.13%
Midmarket	2,381,883	660,103	172,178.0	27.71%
Channel Partners	1,800,594	1,316,803	161,263.5	73.13%

12.2 Country Analysis (from Excel)

Country	Sales	Profit	Margin %
United States of America	25,029,830	2,995,541	11.97%
Canada	24,887,655	3,529,229	14.18%
France	24,354,172	3,781,021	15.53%
Germany	23,505,341	3,680,389	15.66%
Mexico	20,949,352	2,907,523	13.88%

12.3 Product Analysis (from Excel)

Product	Sales	Profit	Margin %
Paseo	33,011,140	4,797,438	14.53%
VTT	20,511,920	3,034,608	14.79%
Amarilla	17,747,121	2,814,104	15.86%
Velo	18,250,059	2,305,992	12.64%
Montana	15,390,819	2,114,755	13.74%
Carretera	13,815,309	1,826,805	13.22%

13. Tableau Visualization

The cleaned sales dataset was connected to Tableau (Tableau Public / Desktop) as an Excel data source. Several individual worksheets were built to analyze different aspects of the business,then combined into a single dashboard named “Sales & Profit Dashboard” (workbook: Sales_Profit_Dashboard.twb).

A. Total Sales

A dedicated worksheet summarizing overall Total Sales across the full dataset, giving a single reference point for the company's total revenue: 118,726,350.

B. Profit by Product

A horizontal bar chart comparing total Profit across the six products. This view makes it easy to see at a glance which products contribute most to profit and which lag behind, based on the Product Analysis figures in Section 12.3.

C. Sales by Segment

A pie chart showing how total sales are distributed across the five customer segments (Government, Small Business, Enterprise, Midmarket, Channel Partners), highlighting each segment's relative share of overall business.

D. Sales by Country

A horizontal bar chart comparing total sales across the five countries (United States of America, Canada, France, Germany, Mexico), showing that sales are fairly evenly spread across markets.

E. Monthly Sales Trend

A line chart plotting total sales by Month Name across the combined dataset, showing month-to-month fluctuation and a clear upward spike toward the later months of the calendar year.

F. Yearly Sales

Yearly sales were analyzed in the Excel workbook (Section 12), showing 2013 at 26,415,256 and 2014 at 92,311,095. Because the dataset only includes September–December for 2013 versus a full 12 months for 2014, this is not a like-for-like year-over-year comparison and should be read as a partial-year figure for 2013.

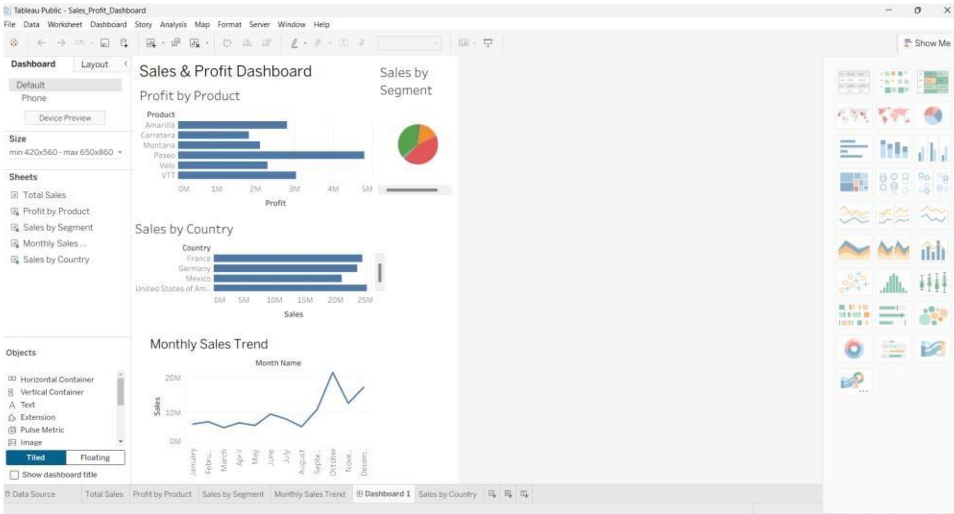

Figure 13.1: Sales & Profit Dashboard in Tableau (actual screenshot)

14. Dashboard Design

The Sales & Profit Dashboard combines four visualizations — Profit by Product, Sales by Segment, Sales by Country, and Monthly Sales Trend — on a single tiled layout, sized to scale between 420x560 and 650x860 pixels for flexible viewing. The layout groups the product- and segment-level views at the top and the country- and time-based views below, so a viewer can move from "what" (products/segments) to "where and when" (country/time) while reading the dashboard top to bottom.

Bar charts were used for Profit by Product and Sales by Country because they make it easy to compare discrete categories side by side. A pie chart was used for Sales by Segment to emphasize each segment's proportional share of total sales. A line chart was used for the Monthly Sales Trend since it is the most effective chart type for showing change over time.

15. Key Business Insights

- Total Sales across the dataset amount to 118,726,350, generating a combined profit of approximately 16,893,702 across all segments.
- Government is the strongest segment, with the highest sales (52,504,261) and highest profit (11,388,173) at a healthy 21.69% margin.
- Enterprise is the only segment operating at a net loss, with total profit of -614,546 (a -3.13% margin) despite generating 19,611,694 in sales — this is the most significant finding of the analysis.
- Channel Partners has by far the highest profit margin (73.13%), even though it is the smallest segment by sales volume, suggesting an efficient but under-scaled part of the business.
- Paseo is the best-performing product by both sales (33,011,140) and profit (4,797,438), clearly ahead of the other five products.
- Carretera is the weakest-performing product by total sales (13,815,309).
- The United States generates the highest country-level sales (25,029,830) but has the lowest profit margin among the five countries (11.97%), while France has the highest country-level margin (15.53%) despite lower total sales than the US or Canada.

- Monthly sales show a clear seasonal pattern, with sales rising notably in the later months of the year based on the Monthly Sales Trend chart and underlying data.

16. Business Recommendations

- Investigate the Enterprise segment's pricing and discounting practices, since it is the only segment generating a net loss despite meaningful sales volume.
- Study what makes Channel Partners so profitable (73.13% margin) and evaluate whether this segment can be scaled without eroding its margin.
- Continue prioritizing the Government segment as the primary driver of both revenue and profit.
- Review the cost and discount structure in the United States market, where sales are highest but margin is the lowest among all five countries.
- Examine the factors behind Paseo's strong performance and assess whether similar positioning or pricing could help lower-performing products such as Carretera.
- Use the observed seasonal sales pattern to plan inventory, staffing, and marketing activity around peak months.
- Treat the 2013 vs 2014 yearly comparison with caution, since 2013 data covers only four months (September–December); a full year of 2013 data (or exclusion of 2013 from year-over-year comparisons) would give a more reliable trend.

17. Challenges and Learning Outcomes

Working through this capstone project provided hands-on experience with a complete analytics workflow rather than an isolated exercise. Some general challenges and learnings from a project of this type typically include:

- Handling missing values (such as the 53 missing Discount Band entries) in a way that preserves the underlying transaction data rather than discarding it.
- Moving analysis results cleanly between tools — from Python/Pandas, to Excel for verification, to Tableau for visualization — while keeping the numbers consistent at every stage.
- Choosing appropriate chart types (bar, pie, line) for different kinds of comparisons, as emphasized in the course's “Storytelling with Data” module.
- Designing a dashboard layout that is easy to read at a glance, balancing multiple charts on one screen.

18. Conclusion

This capstone project successfully demonstrates an end-to-end data analysis workflow: sample business sales dataset in Python, analyzing it with Pandas, validating the results in Excel, and communicating the findings through an interactive Tableau dashboard. The analysis identified clear strengths in the business — particularly the Government segment and the Paseo product — as well as one significant area of concern, the loss-making Enterprise segment, giving stakeholders a concrete, data-backed starting point for decision-making.

19. Future Scope

- Add interactive filters (by segment, country, product, or date) to the Tableau dashboard to let stakeholders explore the data themselves.
- Introduce calculated fields in Tableau (e.g., profit margin %, discount %) directly within the dashboard rather than pre-computing them in Excel.
- Extend the analysis with SQL-based querying if the dataset is moved into a database.
- Collect a full year of 2013 data (or additional years) to enable more reliable year-over-year trend analysis.
- Explore forecasting or predictive analytics as a future extension, once the descriptive analysis foundation established in this project is in place.

20. References

- Dataset: Sample data.xlsx (Sheet1) — company sales dataset used for this project.
- Python Pandas Documentation — https://pandas.pydata.org/docs/
- Tableau Public / Desktop Help — https://help.tableau.com/
- Microsoft Excel Documentation — https://support.microsoft.com/excel
- Axcentra Data Analyst Course — Week 4: Data Visualization & Capstone Project curriculum materials.